

Tan Dai Ngo

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Technical Skills

- **Programming:** Python, Java, SQL, MATLAB
- **Machine Learning:** scikit-learn, XGBoost, CatBoost, regression, classification, clustering, feature engineering, Optuna
- **Data & ML Systems:** Apache Spark, Kafka, Cassandra, Neo4j, MLflow, ETL pipelines, distributed systems
- **Deployment & GenAI:** FastAPI, Docker, AWS (EC2), Sentence Transformers, Qdrant, RAG, Ollama LLM

Education

Master of Engineering in Applied Data Science | GPA: 8.68 / 9.00 *University of Victoria | Sep 2025 - Dec 2027 (Expected)*

- **Coursework:** Optimization for ML, Systems for Massive Datasets, Data Analysis & Pattern Recognition, Data Mining, Algorithms & Data Models, Artificial Intelligence

Master of Science in Computer Science *University of Chicago | Sep 2020 - Mar 2022*

- **Coursework:** Algorithms, Distributed Systems, Advanced Python Programming, Databases, Applied Software Engineering

Bachelor of Science in Economics | GPA: 3.57 / 4.00 *University of Washington | Sep 2015 - Aug 2019*

- **Minors:** Informatics, Applied Mathematics

Experience

Math & Computer Science Teacher | APU American International Schools *Da Nang City, Vietnam | Aug 2024 - Dec 2024*

- Taught Python programming, algorithms, OOP, and software engineering fundamentals to 50+ students through hands-on coding projects with focus on debugging, algorithmic reasoning, and performance optimization.

Software Developer | T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Automated Kafka reprocessing pipelines for 50+ weekly roaming service tests, cutting manual intervention by 80%.
- Redesigned Cassandra schemas for partner/workflow microservices (100k+ records), maintaining query latency under 3s.
- Developed Spring Boot microservices integrated with Jenkins CI/CD and Splunk, enabling scalable downstream analytics.

Projects

Distributed Ocean Data ML Platform with RAG *Systems for Massive Datasets*

- Built an end-to-end distributed ML pipeline using Apache Spark (Bronze → Silver → Gold architecture) to ingest and harmonize 10M+ NOAA and ONC ocean sensor observations for scalable feature engineering and time series forecasting.
- Developed and tuned XGBoost forecasting models using Optuna and MLflow, applying chronological holdout validation on temporally correlated ocean data (RMSE: 0.00586, R²: 0.9999).
- Deployed FastAPI services for model inference and RAG search using Sentence Transformers, Qdrant, and Ollama with retrieval benchmarking (hit@k: 1.0, term recall: 0.875) and a Streamlit dashboard for interactive exploration.

End-to-End Machine Learning System for Multi-Output Fuel Blending *Shell.ai Hackathon 2025*

- Built and deployed a production ML system for multi-output regression across 10 chemical blend properties using FastAPI, Docker, and AWS EC2 for real-time inference.
- Benchmarked XGBoost and CatBoost using 5-fold cross-validation, selecting CatBoost after achieving significantly lower error (MAPE: 0.64 vs. 1.29).
- Designed feature engineering and inference pipelines using weighted aggregation, mixture metrics based on entropy, and serialized preprocessing to ensure consistent production predictions.

Stock Sentiment Analysis (Team of 4) *Algorithms and Data Models*

- Built a sentiment analysis pipeline over 15k+ tweets across 10 tech stocks, enabling large-scale market sentiment analysis, by integrating X API data, SQLite/Neo4j storage, and FinTwitBERT modeling.
- Identified measurable sentiment-return relationships, including statistically significant predictive effects for NVIDIA, by implementing engagement-weighted sentiment aggregation and Granger causality testing.
- Analyzed equity volatility dynamics across major tech stocks, revealing persistent volatility clustering, by applying GARCH(1,1) modeling and automated cross-stock analysis workflows.